By-mode "ECSW"

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October 9, 2026

At $\widetilde{\boldsymbol{u}} = \boldsymbol{u}_{\mathrm{ref}} + \boldsymbol{V}\boldsymbol{q}$, $\boldsymbol{r}_e(\boldsymbol{q})$ contains the assembled HDM residual rows associated with entity e. The local test-basis block is

$$\text{Galerkin:} \quad \boldsymbol{W}_e = \boldsymbol{V}_e, \qquad \text{LSPG:} \quad \boldsymbol{W}_e = \boldsymbol{J}_e \boldsymbol{V}_{e^+}.$$

$\boldsymbol{J}_e = \frac{\partial \boldsymbol{r}_e}{\partial \widetilde{\boldsymbol{u}}_{e^+}}$ is the local residual Jacobian; e^+ contains all state entries needed to evaluate $\boldsymbol{r}_e$.
$\boldsymbol{V}_e$ and $\boldsymbol{V}_{e^+}$ contain the corresponding rows of the trial basis $\boldsymbol{V}$.

$$\text{Full PROM:} \quad \boldsymbol{r}_n(\boldsymbol{q}) = \sum\nolimits_{e \in \mathcal{E}} \boldsymbol{W}_e^T \boldsymbol{r}_e,$$

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$$\text{Classical ECSW:} \quad \widetilde{\boldsymbol{r}}_n^{\mathrm{ECSW}}(\boldsymbol{q}) = \sum\nolimits_{e \in \widetilde{\mathcal{E}}} \xi_e \boldsymbol{W}_e^T \boldsymbol{r}_e,$$

ECSW: one weight per entity.

The sampled mesh and weights are trained offline and fixed online. All residual and test-basis blocks above are evaluated at the current $\boldsymbol{q}$.

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$$\text{BM-ECSW:} \quad \widetilde{\boldsymbol{r}}_n^{\mathrm{BM}}(\boldsymbol{q}) = \sum\nolimits_{e \in \widetilde{\mathcal{E}}} \boldsymbol{D}_e \boldsymbol{W}_e^T \boldsymbol{r}_e.$$

$$\boldsymbol{D}_e = \mathrm{diag}(\xi_{e,1}, \ldots, \xi_{e,n}), \qquad \xi_e, \xi_{e,i} \geq 0, \qquad \widetilde{\mathcal{E}} \subset \mathcal{E}.$$

ECSW: one weight per entity. BM: one weight per entity and reduced component.

The sampled mesh and weights are trained offline and fixed online. All residual and test-basis blocks above are evaluated at the current $\boldsymbol{q}$.

Offline construction: fit the full projected training target using nonnegative weights:

$$\min_{\boldsymbol{\xi}\geq \mathbf{0}} \frac{1}{2}\left\|\boldsymbol{G}\boldsymbol{\xi}-\boldsymbol{b}\right\|_2^2, \qquad \boldsymbol{b}=\boldsymbol{G}\mathbf{1}.$$

Training data: stack the local contributions over N_c configurations:

$$\boldsymbol{G}=\begin{bmatrix} \boldsymbol{g}_1^{(1)} & \cdots & \boldsymbol{g}_{N_e}^{(1)} \\ \vdots & \ddots & \vdots \\ \boldsymbol{g}_1^{(N_c)} & \cdots & \boldsymbol{g}_{N_e}^{(N_c)} \end{bmatrix} \in \mathbb{R}^{nN_c\times N_e}, \qquad \boldsymbol{g}_e^{(c)}=\left[\boldsymbol{W}_e^{(c)}\right]^T \boldsymbol{r}_e^{(c)}.$$

Each column represents one entity; each training configuration supplies n rows.
n: reduced modes; N_e: full-mesh entities.

Progressive sampling: select entities and refit their weights by NNLS until the prescribed accuracy is attained:

$$\frac{\left\|\boldsymbol{G}\boldsymbol{\xi}-\boldsymbol{b}\right\|_2}{\left\|\boldsymbol{b}\right\|_2}\leq \tau, \qquad \widetilde{\mathcal{E}}=\{e:\xi_e>0\}.$$

Classical ECSW: mesh sampling by NNLS

Input: $\boldsymbol{G} = [\boldsymbol{g}_e]$, $\boldsymbol{b} \neq \boldsymbol{0}$, tolerance $\tau \in (0, 1)$
Output: Weights $\boldsymbol{\xi} \geq \boldsymbol{0}$ and reduced mesh $\widetilde{\mathcal{E}}$

1: $k \leftarrow 0$
2: $\mathcal{I}^0 \leftarrow \varnothing$; $\mathcal{Z}^0 \leftarrow \mathcal{E}$
3: $\boldsymbol{\xi}^0 \leftarrow \boldsymbol{0}$
4: $\boldsymbol{h}^0 \leftarrow \boldsymbol{b}$
5: **while** $\|\boldsymbol{h}^k\|_2 > \tau \|\boldsymbol{b}\|_2$ **do**
6: Compute the correlations: $\boldsymbol{\mu}^k \leftarrow \boldsymbol{G}^T \boldsymbol{h}^k$
7: Select the most correlated entity: $e^k \leftarrow \arg\max_{e \in \mathcal{Z}^k} \mu_e^k$
8: Add the candidate: $\mathcal{I}^{k+1} \leftarrow \mathcal{I}^k \cup \{e^k\}$
9: Update the unselected set: $\mathcal{Z}^{k+1} \leftarrow \mathcal{E} \setminus \mathcal{I}^{k+1}$
10: Refit the weights of all selected entities by NNLS:

$$\boldsymbol{\xi}_{\mathcal{I}^{k+1}}^{k+1} \in \arg\min_{\boldsymbol{z} \geq \boldsymbol{0}} \frac{1}{2} \|\boldsymbol{G}_{\mathcal{I}^{k+1}} \boldsymbol{z} - \boldsymbol{b}\|_2^2$$

11: Set unselected weights to zero: $\boldsymbol{\xi}_{\mathcal{Z}^{k+1}}^{k+1} \leftarrow \boldsymbol{0}$
12: Recompute the full fitting residual: $\boldsymbol{h}^{k+1} \leftarrow \boldsymbol{b} - \boldsymbol{G}\boldsymbol{\xi}^{k+1}$
13: $k \leftarrow k + 1$
14: **end while**
15: $\widetilde{\mathcal{E}} \leftarrow \{e \in \mathcal{E} : \xi_e^k > 0\}$
16: **return** $\boldsymbol{\xi}^k$ and $\widetilde{\mathcal{E}}$

Click a red line number for its explanation.

Offline construction: fit the projected contributions with one weight vector per mode:

$$\min_{\boldsymbol{\xi}^{[i]} \geq \mathbf{0}} \frac{1}{2} \| \boldsymbol{G}^{[i]} \boldsymbol{\xi}^{[i]} - \boldsymbol{b}^{[i]} \|_2^2, \qquad \boldsymbol{\xi}^{[i]} = [\xi_{1,i}, \ldots, \xi_{N_e,i}]^T.$$

Same data, grouped by mode: collect the rows of $\boldsymbol{G}$ associated with reduced equation i, one per training configuration:

$$\boldsymbol{G}^{[i]} = \begin{bmatrix} g_{1,i}^{(1)} & \cdots & g_{N_e,i}^{(1)} \\ \vdots & \ddots & \vdots \\ g_{1,i}^{(N_c)} & \cdots & g_{N_e,i}^{(N_c)} \end{bmatrix} \in \mathbb{R}^{N_c \times N_e}, \qquad g_{e,i}^{(c)} = [\boldsymbol{W}_e^{(c)}(:,i)]^T \boldsymbol{r}_e^{(c)},$$

$$\boldsymbol{b}^{[i]} = \boldsymbol{G}^{[i]} \mathbf{1}.$$

e: entity; i: reduced equation; c: training configuration.

One shared mesh, one accuracy target: combine the modal fitting errors:

$$\boldsymbol{h}^{[i]} = \boldsymbol{b}^{[i]} - \boldsymbol{G}^{[i]} \boldsymbol{\xi}^{[i]}, \qquad \eta^2 = \frac{\sum_i \| \boldsymbol{h}^{[i]} \|_2^2}{\sum_i \| \boldsymbol{b}^{[i]} \|_2^2} \leq \tau^2.$$

Entities are selected jointly; on the selected set, each mode's weights are refitted independently.

Input: $\boldsymbol{G}^{[i]} = [\boldsymbol{g}_e^{[i]}]$, $\boldsymbol{b}^{[i]}$ with $\sum_i \|\boldsymbol{b}^{[i]}\|_2^2 > 0$, tolerance $\tau \in (0, 1)$

Output: Modal weights $\boldsymbol{\xi}^{[i]} \geq \boldsymbol{0}$ and reduced mesh $\widetilde{\mathcal{E}}$

1: $k \leftarrow 0$; $\mathcal{I}^0 \leftarrow \varnothing$; $\mathcal{Z}^0 \leftarrow \mathcal{E}$

2: $\boldsymbol{\xi}^{[i],0} \leftarrow \boldsymbol{0}$; $\boldsymbol{h}^{[i],0} \leftarrow \boldsymbol{b}^{[i]}$ for every i

3: **while** $\sum_i \|\boldsymbol{h}^{[i],k}\|_2^2 > \tau^2 \sum_i \|\boldsymbol{b}^{[i]}\|_2^2$ **do**

4: Sum the modal correlations: $\boldsymbol{\mu}^k \leftarrow \sum_{i=1}^n [\boldsymbol{G}^{[i]}]^T \boldsymbol{h}^{[i],k}$

5: Select the most correlated entity: $e^k \leftarrow \arg\max_{e \in \mathcal{Z}^k} \mu_e^k$

6: Add the candidate: $\mathcal{I}^{k+1} \leftarrow \mathcal{I}^k \cup \{e^k\}$

7: Update the unselected set: $\mathcal{Z}^{k+1} \leftarrow \mathcal{E} \setminus \mathcal{I}^{k+1}$

8: **for** $i = 1, \ldots, n$ **do**

9: Refit the selected weights of mode i by NNLS:

$$\boldsymbol{\xi}^{[i],k+1}_{\mathcal{I}^{k+1}} \in \arg\min_{\boldsymbol{z} \geq \boldsymbol{0}} \frac{1}{2} \|\boldsymbol{G}^{[i]}_{\mathcal{I}^{k+1}} \boldsymbol{z} - \boldsymbol{b}^{[i]}\|_2^2$$

10: Set unselected weights to zero: $\boldsymbol{\xi}^{[i],k+1}_{\mathcal{Z}^{k+1}} \leftarrow \boldsymbol{0}$

11: Recompute the fitting residual: $\boldsymbol{h}^{[i],k+1} \leftarrow \boldsymbol{b}^{[i]} - \boldsymbol{G}^{[i]} \boldsymbol{\xi}^{[i],k+1}$

12: **end for**

13: $k \leftarrow k + 1$

14: **end while**

15: $\widetilde{\mathcal{E}} \leftarrow \bigcup_{i=1}^n \{e \in \mathcal{E} : \xi^k_{e,i} > 0\}$

16: **return** $\{\boldsymbol{\xi}^{[i],k}\}_{i=1}^n$ and $\widetilde{\mathcal{E}}$

Online target: solve $\widetilde{\boldsymbol{r}}_n(\boldsymbol{q}) = \boldsymbol{0}$ at each time step, with the trained mesh and weights fixed.
At iteration k: $\boldsymbol{r}_e^k = \boldsymbol{r}_e(\boldsymbol{q}^k)$ and $\boldsymbol{W}_e^k = \frac{\partial \boldsymbol{r}_e}{\partial \boldsymbol{q}}(\boldsymbol{q}^k)$.

HPROM (ECSW)

Sampled projected residual:

$$\widetilde{\boldsymbol{r}}_n^{\mathrm{ECSW},k} = \sum_{e \in \widetilde{\mathcal{E}}} \xi_e [\boldsymbol{W}_e^k]^T \boldsymbol{r}_e^k.$$

Gauss–Newton Jacobian approximation:

$$\widehat{\boldsymbol{J}}_n^{\mathrm{ECSW},k} = \sum_{e \in \widetilde{\mathcal{E}}} \xi_e [\boldsymbol{W}_e^k]^T \boldsymbol{W}_e^k.$$

Coupled correction:

$$\widehat{\boldsymbol{J}}_n^{\mathrm{ECSW},k} \Delta \boldsymbol{q}^k = -\widetilde{\boldsymbol{r}}_n^{\mathrm{ECSW},k}.$$

Classical online algorithm

HPROM (BM-ECSW)

Sampled projected residual:

$$\widetilde{\boldsymbol{r}}_n^{\mathrm{BM},k} = \sum_{e \in \widetilde{\mathcal{E}}} \boldsymbol{D}_e [\boldsymbol{W}_e^k]^T \boldsymbol{r}_e^k.$$

Gauss–Newton-type Jacobian approximation:

$$\widehat{\boldsymbol{J}}_n^{\mathrm{BM},k} = \sum_{e \in \widetilde{\mathcal{E}}} \boldsymbol{D}_e [\boldsymbol{W}_e^k]^T \boldsymbol{W}_e^k.$$

Coupled correction:

$$\widehat{\boldsymbol{J}}_n^{\mathrm{BM},k} \Delta \boldsymbol{q}^k = -\widetilde{\boldsymbol{r}}_n^{\mathrm{BM},k}.$$

BM online algorithm

Update all coordinates together: $\boldsymbol{q}^{k+1} = \boldsymbol{q}^k + \Delta \boldsymbol{q}^k$.

Correction matrices: classical, symmetric positive semidefinite; BM, possibly nonsymmetric.

2D parametric inviscid Burgers; same ROB and training data ($N_e = 62\,500$, $n = 96$, $N_c = 225$, $\tau = 10^{-5}$).

Offline

Computational model	n	n_e	n_e^+	η	Offline training time (min)
HPROM (ECSW)	96	2 492	5 914	9.795×10^{-6}	112.56
HPROM (BM-ECSW)	96	300	868	8.359×10^{-6}	4.10

n_e^+: augmented-mesh cells. Offline excludes HDM generation and POD; speedups use unrounded times.

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Computational model	n (N for HDM)	n_e (N_e for HDM)	n_e^+	$\mathbb{RE}_{2,\boldsymbol{u}}^{\max}$ (%)	Wall-clock time (min)	Speedup factor
HDM	125 000	62 500	62 500	–	5.404	–
HPROM (ECSW)	96	2 492	5 914	1.096	0.311	17.39
HPROM (BM-ECSW)	96	300	868	1.102	0.020	267.93

8.31× fewer cells; 27.48× faster training; 15.4× lower online time; comparable accuracy.

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With $N_c = 90$: 145 cells, but 4 411.533% maximum trajectory error.

Goal: preserve $\boldsymbol{F} = \nabla_{\boldsymbol{q}}\Phi$ after weighting, so that LSPG retains one common error objective and structural dynamics retains one common potential.

$\boldsymbol{F}$: LSPG projected residual; reduced structural internal vector obtained by Galerkin projection.

Smooth local scalar for entity e: LSPG: $\phi_e = \frac{1}{2}\|\boldsymbol{r}_e(\boldsymbol{q})\|_2^2$; structures: $\phi_e = \mathcal{V}_{e,r}^{\mathrm{int}}(\boldsymbol{q})$.

$$\boldsymbol{F}(\boldsymbol{q}) = \sum\nolimits_{e\in\mathcal{E}} \nabla_{\boldsymbol{q}}\phi_e(\boldsymbol{q}).$$

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$$\boldsymbol{F}(\boldsymbol{q}) = \sum\nolimits_{e\in\mathcal{E}} \nabla_{\boldsymbol{q}}\phi_e(\boldsymbol{q}).$$

Now we can connect these local functions to one common scalar function.

$$\Phi(\boldsymbol{q}) = \sum\nolimits_{e\in\mathcal{E}} \phi_e(\boldsymbol{q}), \qquad \boxed{\boldsymbol{F} = \nabla_{\boldsymbol{q}}\Phi}, \qquad F_i = \frac{\partial\Phi}{\partial q_i}, \qquad i = 1,\ldots,n.$$

Goal: preserve $\boldsymbol{F} = \nabla_{\boldsymbol{q}}\Phi$ after weighting, so that LSPG retains one common error objective and structural dynamics retains one common potential.

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The question is whether the weighted vector is still the gradient of one common scalar function.

If $\boldsymbol{F} = \nabla_{\boldsymbol{q}}\Phi$, differentiating F_i in q_j differentiates Φ first in q_i, then in q_j.
Differentiating F_j in q_i reverses the order. Continuous second derivatives give the same result:

$$\frac{\partial F_i}{\partial q_j} = \frac{\partial^2\Phi}{\partial q_j \partial q_i} = \frac{\partial^2\Phi}{\partial q_i \partial q_j} = \frac{\partial F_j}{\partial q_i}.$$

Goal: preserve $\boldsymbol{F} = \nabla_{\boldsymbol{q}}\Phi$ after weighting, so that LSPG retains one common error objective and structural dynamics retains one common potential.

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Preserving this connection allows us to retain the following classical properties:

- **LSPG:** the classical Gauss–Newton correction minimizes one common quadratic error.
- **Structural dynamics:** one common potential defines the Lagrangian and conserved energy.

Structures: same reduced kinetic energy, continuous time. BM may still converge without this structure.

The sampled mesh $\widetilde{\mathcal{E}}$ and the trained weights are fixed.

First, apply classical ECSW: one weight per entity, shared by all components.

$$\boldsymbol{F}_{\mathrm{ECSW}} = \sum\nolimits_{e \in \widetilde{\mathcal{E}}} \xi_e \nabla_{\boldsymbol{q}} \phi_e.$$

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The same weight keeps each pair equal and preserves one common scalar function.

$$\frac{\partial F_{\mathrm{ECSW},i}}{\partial q_j} - \frac{\partial F_{\mathrm{ECSW},j}}{\partial q_i} = \sum\nolimits_{e\in\widetilde{\mathcal{E}}} \xi_e \left[\frac{\partial^2 \phi_e}{\partial q_j \partial q_i} - \frac{\partial^2 \phi_e}{\partial q_i \partial q_j} \right] = 0.$$

$$\Phi_{\mathrm{ECSW}} = \sum\nolimits_{e\in\widetilde{\mathcal{E}}} \xi_e \phi_e, \qquad \nabla_{\boldsymbol{q}} \Phi_{\mathrm{ECSW}} = \sum\nolimits_{e\in\widetilde{\mathcal{E}}} \xi_e \nabla_{\boldsymbol{q}} \phi_e = \boldsymbol{F}_{\mathrm{ECSW}}.$$

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Now apply BM-ECSW: components of one entity can receive different weights.

$$\boldsymbol{F}_{\mathrm{BM}} = \sum\nolimits_{e\in\widetilde{\mathcal{E}}} \boldsymbol{D}_e \nabla_{\boldsymbol{q}} \phi_e, \qquad F_{\mathrm{BM},i} = \sum\nolimits_{e\in\widetilde{\mathcal{E}}} \xi_{e,i} \frac{\partial \phi_e}{\partial q_i}.$$

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$$\Phi_{\mathrm{ECSW}} = \sum\nolimits_{e\in\widetilde{\mathcal{E}}} \xi_e \phi_e, \qquad \nabla_{\boldsymbol{q}} \Phi_{\mathrm{ECSW}} = \sum\nolimits_{e\in\widetilde{\mathcal{E}}} \xi_e \nabla_{\boldsymbol{q}} \phi_e = \boldsymbol{F}_{\mathrm{ECSW}}.$$

Now apply BM-ECSW: components of one entity can receive different weights.

$$\boldsymbol{F}_{\mathrm{BM}} = \sum\nolimits_{e\in\widetilde{\mathcal{E}}} \boldsymbol{D}_e \nabla_{\boldsymbol{q}} \phi_e, \qquad F_{\mathrm{BM},i} = \sum\nolimits_{e\in\widetilde{\mathcal{E}}} \xi_{e,i} \frac{\partial \phi_e}{\partial q_i}.$$

The local derivatives agree, but they can be multiplied by different weights.

$$\frac{\partial F_{\mathrm{BM},i}}{\partial q_j} - \frac{\partial F_{\mathrm{BM},j}}{\partial q_i} = \underbrace{\sum\nolimits_{e\in\widetilde{\mathcal{E}}} [\xi_{e,i} - \xi_{e,j}] \frac{\partial^2 \phi_e}{\partial q_i \partial q_j}}_{\text{not necessarily zero}}.$$

Therefore, BM-ECSW no longer guarantees a common error objective or potential.

A nonzero difference excludes every common smooth Φ_{BM} locally.

Compatible cases: shared weights, zero local mixed derivatives, or cancellation.

Full differentiation

Galerkin projection gives the reduced kinetic energy and element potential energies:

$$\mathcal{T}_r = \tfrac{1}{2}\dot{\boldsymbol{q}}^T \boldsymbol{M}_r \dot{\boldsymbol{q}}, \qquad \phi_e(\boldsymbol{q}) = \mathcal{V}^{\text{int}}_{e,r}(\boldsymbol{q}).$$

$\boldsymbol{M}_r = \boldsymbol{V}^T \boldsymbol{M} \boldsymbol{V}$ is constant, symmetric positive definite.

Classical ECSW retains one common potential for the reduced mesh:

$$\Phi_{\text{ECSW}} = \sum\nolimits_{e \in \widetilde{\mathcal{E}}} \xi_e \phi_e, \qquad \nabla_{\boldsymbol{q}} \Phi_{\text{ECSW}} = \sum\nolimits_{e \in \widetilde{\mathcal{E}}} \xi_e \nabla_{\boldsymbol{q}} \phi_e = \boldsymbol{F}_{\text{ECSW}}.$$

Same reduced kinetic energy; time-independent potentials; no damping or external loading.
Conservation in continuous time. Farhat et al. (2015), Section 4.1.

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1. Preservation of the Lagrangian structure

$$\widetilde{\mathcal{L}}_r^{\text{ECSW}} = \underbrace{\mathcal{T}_r}_{\text{kinetic energy}} - \underbrace{\Phi_{\text{ECSW}}}_{\text{potential energy}}.$$

Euler–Lagrange gives the equations of motion:

$$\frac{d}{dt}\left[\frac{\partial \widetilde{\mathcal{L}}_r^{\text{ECSW}}}{\partial \dot{\boldsymbol{q}}}\right] - \frac{\partial \widetilde{\mathcal{L}}_r^{\text{ECSW}}}{\partial \boldsymbol{q}} = \boldsymbol{M}_r \ddot{\boldsymbol{q}} + \boldsymbol{F}_{\text{ECSW}} = \boldsymbol{0}.$$

Euler–Lagrange steps

Same reduced kinetic energy; time-independent potentials; no damping or external loading.

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Galerkin projection gives the reduced kinetic energy and element potential energies:

$$\mathcal{T}_r = \tfrac{1}{2}\dot{\boldsymbol{q}}^T \boldsymbol{M}_r \dot{\boldsymbol{q}}, \qquad \phi_e(\boldsymbol{q}) = \mathcal{V}^{\text{int}}_{e,r}(\boldsymbol{q}).$$

$\boldsymbol{M}_r = \boldsymbol{V}^T \boldsymbol{M} \boldsymbol{V}$ is constant, symmetric positive definite.

Classical ECSW retains one common potential for the reduced mesh:

$$\Phi_{\text{ECSW}} = \sum\nolimits_{e\in\widetilde{\mathcal{E}}} \xi_e \phi_e, \qquad \nabla_{\boldsymbol{q}} \Phi_{\text{ECSW}} = \sum\nolimits_{e\in\widetilde{\mathcal{E}}} \xi_e \nabla_{\boldsymbol{q}} \phi_e = \boldsymbol{F}_{\text{ECSW}}.$$

1. Preservation of the Lagrangian structure

$$\widetilde{\mathcal{L}}^{\text{ECSW}}_r = \underbrace{\mathcal{T}_r}_{\text{kinetic energy}} - \underbrace{\Phi_{\text{ECSW}}}_{\text{potential energy}}.$$

Euler–Lagrange gives the equations of motion:

$$\frac{d}{dt}\left[\frac{\partial \widetilde{\mathcal{L}}^{\text{ECSW}}_r}{\partial \dot{\boldsymbol{q}}}\right] - \frac{\partial \widetilde{\mathcal{L}}^{\text{ECSW}}_r}{\partial \boldsymbol{q}} = \boldsymbol{M}_r \ddot{\boldsymbol{q}} + \boldsymbol{F}_{\text{ECSW}} = \boldsymbol{0}.$$

Euler–Lagrange steps

2. Conservation of total mechanical energy

$$\widetilde{\mathcal{H}}_r = \mathcal{T}_r + \Phi_{\text{ECSW}}.$$

Along these equations of motion,

$$\frac{d\widetilde{\mathcal{H}}_r}{dt} = \dot{\boldsymbol{q}}^T[\boldsymbol{M}_r \ddot{\boldsymbol{q}} + \nabla_{\boldsymbol{q}} \Phi_{\text{ECSW}}] = \dot{\boldsymbol{q}}^T[-\boldsymbol{F}_{\text{ECSW}} + \boldsymbol{F}_{\text{ECSW}}] = 0.$$

Kinetic and potential energy can change, while their sum remains constant.

Same reduced kinetic energy; time-independent potentials; no damping or external loading.

Conservation in continuous time. Farhat et al. (2015), Section 4.1.

Full derivation

BM keeps the same kinetic energy but changes the internal vector and equations of motion:

$$\boldsymbol{F}_{\mathrm{BM}} = \sum\nolimits_{e \in \widetilde{\mathcal{E}}} \boldsymbol{D}_e \nabla_{\boldsymbol{q}} \mathcal{V}_{e,r}^{\mathrm{int}}, \qquad \boldsymbol{M}_r \ddot{\boldsymbol{q}} + \boldsymbol{F}_{\mathrm{BM}} = \boldsymbol{0}.$$

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What must a potential Φ_{BM} satisfy to preserve the two properties of Slide 11?

1. Lagrangian structure

$$\widetilde{\mathcal{L}}_r^{\mathrm{BM}} = \mathcal{T}_r - \Phi_{\mathrm{BM}}.$$

Euler–Lagrange would give

$$\boldsymbol{M}_r \ddot{\boldsymbol{q}} + \nabla_{\boldsymbol{q}} \Phi_{\mathrm{BM}} = \boldsymbol{0}.$$

Its gradient must reproduce the BM internal vector.

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2. Total mechanical energy

$$\widetilde{\mathcal{H}}^{\mathrm{BM}}_r = \mathcal{T}_r + \Phi_{\mathrm{BM}}.$$

Along the actual BM equations,

$$\frac{d\widetilde{\mathcal{H}}^{\mathrm{BM}}_r}{dt} = \dot{\boldsymbol{q}}^T [\nabla_{\boldsymbol{q}} \Phi_{\mathrm{BM}} - \boldsymbol{F}_{\mathrm{BM}}].$$

Conservation for all motions requires the same identity.

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Conservation for all motions requires the same identity.

Both properties require the same gradient identity:

$$\boxed{\nabla_{\boldsymbol{q}} \Phi_{\mathrm{BM}} = \boldsymbol{F}_{\mathrm{BM}}}.$$

The crossed-derivative test of Slide 10 checks this identity. A nonzero difference excludes every common smooth potential locally.

BM no longer guarantees the inherited Lagrangian structure or conservation of total mechanical energy with the same reduced kinetic energy.

Continuous time. Compatible BM cases and alternative mechanical formulations remain possible.

Crossed-derivative test | Structural counterexample

In LSPG, each local scalar measures an entity's residual error.

$$\phi_e(\boldsymbol{q}) = \tfrac{1}{2}\|\boldsymbol{r}_e(\boldsymbol{q})\|_2^2, \qquad \boldsymbol{W}_e = \frac{\partial \boldsymbol{r}_e}{\partial \boldsymbol{q}}, \qquad \nabla_{\boldsymbol{q}}\phi_e = \boldsymbol{W}_e^T \boldsymbol{r}_e.$$

These local gradients assemble the full LSPG projected residual:

$$\boldsymbol{F} = \sum\nolimits_{e\in\mathcal{E}} \nabla_{\boldsymbol{q}}\phi_e = \sum\nolimits_{e\in\mathcal{E}} \boldsymbol{W}_e^T \boldsymbol{r}_e = \boldsymbol{r}_n.$$

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Classical ECSW preserves this connection after sampling.

$$\Phi_{\mathrm{ECSW}} = \sum\nolimits_{e\in\widetilde{\mathcal{E}}} \xi_e \phi_e = \tfrac{1}{2}\sum\nolimits_{e\in\widetilde{\mathcal{E}}} \xi_e \|\boldsymbol{r}_e\|_2^2,$$

$$\nabla_{\boldsymbol{q}}\Phi_{\mathrm{ECSW}} = \sum\nolimits_{e\in\widetilde{\mathcal{E}}} \xi_e \nabla_{\boldsymbol{q}}\phi_e = \sum\nolimits_{e\in\widetilde{\mathcal{E}}} \xi_e \boldsymbol{W}_e^T \boldsymbol{r}_e = \widetilde{\boldsymbol{r}}_n^{\mathrm{ECSW}} = \boldsymbol{F}_{\mathrm{ECSW}}.$$

All projected equations express stationarity of the same residual-error measure:

$$\widetilde{\boldsymbol{r}}_n^{\mathrm{ECSW}} = \boldsymbol{0} \quad \Longleftrightarrow \quad \nabla_{\boldsymbol{q}}\Phi_{\mathrm{ECSW}} = \boldsymbol{0}.$$

The sampled objective generally differs from the full LSPG objective $\Phi = \sum_{e\in\mathcal{E}} \phi_e$.

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The sampled objective generally differs from the full LSPG objective $\Phi = \sum_{e\in\mathcal{E}} \phi_e$.

BM changes the projected vector by weighting its components separately:

$$\boldsymbol{F}_{\mathrm{BM}} = \widetilde{\boldsymbol{r}}_n^{\mathrm{BM}} = \sum\nolimits_{e\in\widetilde{\mathcal{E}}} \boldsymbol{D}_e \nabla_{\boldsymbol{q}}\phi_e = \sum\nolimits_{e\in\widetilde{\mathcal{E}}} \boldsymbol{D}_e \boldsymbol{W}_e^T \boldsymbol{r}_e.$$

Does one scalar residual-error measure still generate all BM equations?

Next: change one modal weight and check the crossed derivatives.

Full differentiation

LSPG example: changing one modal weight

The same local residuals and scalar functions are used in both rules:

$$r_1 = q_1, \qquad r_2 = q_2, \qquad r_3 = q_1 + q_2^2 - 3, \qquad \text{where } \phi_e = \tfrac{1}{2}\|\boldsymbol{r}_e\|_2^2.$$

Keep all three entities and change only the second weight of entity 3.

Classical ECSW $\boldsymbol{\xi} = (2, 3, 2)^T$.

BM-ECSW $\boldsymbol{\Xi} = \begin{pmatrix} 2 & 2 \\ 3 & 3 \\ 2 & \mathbf{1} \end{pmatrix}$.

Rows of $\boldsymbol{\Xi}$ identify entities; columns identify reduced equations.

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The projected vectors are $\boldsymbol{F}_{\mathrm{ECSW}} = \widetilde{\boldsymbol{r}}_n^{\mathrm{ECSW}}$ and $\boldsymbol{F}_{\mathrm{BM}} = \widetilde{\boldsymbol{r}}_n^{\mathrm{BM}}$:

$$\widetilde{\boldsymbol{r}}_n^{\mathrm{ECSW}} = \sum_{e=1}^{3} \xi_e \nabla_{\boldsymbol{q}} \phi_e = \begin{pmatrix} 2q_1 + 2r_3 \\ 3q_2 + 4q_2 r_3 \end{pmatrix}. \qquad \widetilde{\boldsymbol{r}}_n^{\mathrm{BM}} = \sum_{e=1}^{3} \boldsymbol{D}_e \nabla_{\boldsymbol{q}} \phi_e = \begin{pmatrix} 2q_1 + 2r_3 \\ 3q_2 + 2q_2 r_3 \end{pmatrix}.$$

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Now check whether one scalar function could generate each complete vector.

$$\frac{\partial \widetilde{r}_{n,1}^{\mathrm{ECSW}}}{\partial q_2} = 2\frac{\partial r_3}{\partial q_2} = 4q_2, \qquad \frac{\partial \widetilde{r}_{n,1}^{\mathrm{BM}}}{\partial q_2} = 2\frac{\partial r_3}{\partial q_2} = 4q_2,$$

$$\frac{\partial \widetilde{r}_{n,2}^{\mathrm{ECSW}}}{\partial q_1} = 4q_2\frac{\partial r_3}{\partial q_1} = 4q_2. \qquad \frac{\partial \widetilde{r}_{n,2}^{\mathrm{BM}}}{\partial q_1} = 2q_2\frac{\partial r_3}{\partial q_1} = 2q_2.$$

The same local residuals and scalar functions are used in both rules:

$$r_1 = q_1, \qquad r_2 = q_2, \qquad r_3 = q_1 + q_2^2 - 3, \qquad \text{where } \phi_e = \tfrac{1}{2}\|\boldsymbol{r}_e\|_2^2.$$

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Full calculation

The projected vectors are $\boldsymbol{F}_{\mathrm{ECSW}} = \widetilde{\boldsymbol{r}}_n^{\mathrm{ECSW}}$ and $\boldsymbol{F}_{\mathrm{BM}} = \widetilde{\boldsymbol{r}}_n^{\mathrm{BM}}$:

$$\widetilde{\boldsymbol{r}}_n^{\mathrm{ECSW}} = \sum_{e=1}^{3} \xi_e \nabla_{\boldsymbol{q}} \phi_e = \begin{pmatrix} 2q_1 + 2r_3 \\ 3q_2 + 4q_2 r_3 \end{pmatrix}. \qquad \widetilde{\boldsymbol{r}}_n^{\mathrm{BM}} = \sum_{e=1}^{3} \boldsymbol{D}_e \nabla_{\boldsymbol{q}} \phi_e = \begin{pmatrix} 2q_1 + 2r_3 \\ 3q_2 + 2q_2 r_3 \end{pmatrix}.$$

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$$\boxed{4q_2 = 4q_2} \text{ for every } \boldsymbol{q}.$$
$$\Phi_{\mathrm{ECSW}} = \textstyle\sum_{e=1}^{3} \xi_e \phi_e = \frac{1}{2}\sum_{e=1}^{3} \xi_e \|\boldsymbol{r}_e\|_2^2.$$

$$\frac{\partial \widetilde{r}_{n,1}^{\mathrm{BM}}}{\partial q_2} = 2\frac{\partial r_3}{\partial q_2} = 4q_2,$$
$$\frac{\partial \widetilde{r}_{n,2}^{\mathrm{BM}}}{\partial q_1} = 2q_2\frac{\partial r_3}{\partial q_1} = 2q_2.$$
$$\boxed{4q_2 \neq 2q_2} \text{ when } q_2 \neq 0.$$
No common Φ_{BM} exists locally where $q_2 \neq 0$.

Changing one positive weight is enough to lose the common objective.

Both methods start from the same local linearization:

$$\boldsymbol{r}_e(\boldsymbol{q} + \Delta\boldsymbol{q}) \approx \boldsymbol{r}_e + \boldsymbol{W}_e \Delta\boldsymbol{q}, \qquad \boldsymbol{W}_e = \frac{\partial \boldsymbol{r}_e}{\partial \boldsymbol{q}}.$$

Evaluate $\boldsymbol{r}_e$ and $\boldsymbol{W}_e$ at the current iterate; keep $\boldsymbol{W}_e$ and the trained weights fixed.

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Classical ECSW: Gauss–Newton

Classical ECSW has a common objective.
Substitute the linearized residuals and minimize
its quadratic model:

$$\min_{\Delta\boldsymbol{q}} \ \tfrac{1}{2} \sum\nolimits_{e \in \widetilde{\mathcal{E}}} \xi_e \|\boldsymbol{r}_e + \boldsymbol{W}_e \Delta\boldsymbol{q}\|_2^2.$$

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Differentiate in $\Delta\boldsymbol{q}$ and set to zero:

$$\sum\nolimits_{e\in\widetilde{\mathcal{E}}} \xi_e \boldsymbol{W}_e^T[\boldsymbol{r}_e + \boldsymbol{W}_e\Delta\boldsymbol{q}] = \boldsymbol{0}.$$

$$\widehat{\boldsymbol{J}}_n^{\mathrm{ECSW}}\Delta\boldsymbol{q}^{\mathrm{ECSW}} = -\widetilde{\boldsymbol{r}}_n^{\mathrm{ECSW}}.$$

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BM-ECSW: weighted equations

A common objective is not guaranteed. Instead, linearize and solve the BM projected equations:

$$\sum\nolimits_{e\in\widetilde{\mathcal{E}}} \boldsymbol{D}_e\boldsymbol{W}_e^T[\boldsymbol{r}_e + \boldsymbol{W}_e\Delta\boldsymbol{q}] = \boldsymbol{0}.$$

Expanding gives the BM online system:

$$\widehat{\boldsymbol{J}}_n^{\mathrm{BM}}\Delta\boldsymbol{q}^{\mathrm{BM}} = -\widetilde{\boldsymbol{r}}_n^{\mathrm{BM}}.$$

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Expanding gives the BM online system:

$$\widehat{\boldsymbol{J}}_n^{\mathrm{BM}}\Delta\boldsymbol{q}^{\mathrm{BM}} = -\widetilde{\boldsymbol{r}}_n^{\mathrm{BM}}.$$

- **Step control:** the classical justification that a sufficiently small step decreases the sampled objective no longer applies automatically.
- **Online convergence:** stagnation, oscillations, or divergence are practical risks to check, even when the training fit is accurate.

BM no longer inherits the classical objective-based descent argument.

BM can still converge. The computations assess these risks; the gradient test alone does not predict the outcome.

Online systems

Now examine the matrices used to compute the online correction.

The quadratic form is $z^T \widehat{J}_n z$, $z \in \mathbb{R}^n$.

Classical ECSW

$$\widehat{J}_n^{\mathrm{ECSW}} = \sum_{e \in \widetilde{\mathcal{E}}} \xi_e W_e^T W_e.$$

BM-ECSW

$$\widehat{J}_n^{\mathrm{BM}} = \sum_{e \in \widetilde{\mathcal{E}}} D_e W_e^T W_e.$$

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$$\widehat{J}_n^{\mathrm{ECSW}} = \sum_{e \in \widetilde{\mathcal{E}}} \xi_e W_e^T W_e.$$

Scalar weights preserve symmetry:

$$[\widehat{J}_n^{\mathrm{ECSW}}]^T = \sum_{e \in \widetilde{\mathcal{E}}} \xi_e [W_e^T W_e]^T = \widehat{J}_n^{\mathrm{ECSW}}.$$

Its quadratic form is a sum of squares:

$$z^T \widehat{J}_n^{\mathrm{ECSW}} z = \sum_{e \in \widetilde{\mathcal{E}}} \xi_e \| W_e z \|_2^2 \geq 0.$$

Symmetric positive semidefinite for $\xi_e \geq 0$.

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Symmetric positive semidefinite for $\xi_e \geq 0$.

BM-ECSW

$$\widehat{J}_n^{\mathrm{BM}} = \sum_{e \in \widetilde{\mathcal{E}}} D_e W_e^T W_e.$$

Transposition moves D_e to the right:

$$[D_e W_e^T W_e]^T = W_e^T W_e D_e.$$

The transpose can differ from the original.

$$z^T \widehat{J}_n^{\mathrm{BM}} z = \sum_{e \in \widetilde{\mathcal{E}}} [W_e D_e z]^T [W_e z].$$

These dot products involve different vectors and can be negative.

Now examine the matrices used to compute the online correction.

The quadratic form is $\boldsymbol{z}^T \widehat{\boldsymbol{J}}_n \boldsymbol{z}$, $\boldsymbol{z} \in \mathbb{R}^n$.

Classical ECSW

$$\widehat{\boldsymbol{J}}_n^{\mathrm{ECSW}} = \sum_{e \in \widetilde{\mathcal{E}}} \xi_e \boldsymbol{W}_e^T \boldsymbol{W}_e.$$

Scalar weights preserve symmetry:

$$[\widehat{\boldsymbol{J}}_n^{\mathrm{ECSW}}]^T = \sum_{e \in \widetilde{\mathcal{E}}} \xi_e [\boldsymbol{W}_e^T \boldsymbol{W}_e]^T = \widehat{\boldsymbol{J}}_n^{\mathrm{ECSW}}.$$

Its quadratic form is a sum of squares:

$$\boldsymbol{z}^T \widehat{\boldsymbol{J}}_n^{\mathrm{ECSW}} \boldsymbol{z} = \sum_{e \in \widetilde{\mathcal{E}}} \xi_e \|\boldsymbol{W}_e \boldsymbol{z}\|_2^2 \geq 0.$$

Symmetric positive semidefinite for $\xi_e \geq 0$.

BM-ECSW

$$\widehat{\boldsymbol{J}}_n^{\mathrm{BM}} = \sum_{e \in \widetilde{\mathcal{E}}} \boldsymbol{D}_e \boldsymbol{W}_e^T \boldsymbol{W}_e.$$

Transposition moves $\boldsymbol{D}_e$ to the right:

$$[\boldsymbol{D}_e \boldsymbol{W}_e^T \boldsymbol{W}_e]^T = \boldsymbol{W}_e^T \boldsymbol{W}_e \boldsymbol{D}_e.$$

The transpose can differ from the original.

$$\boldsymbol{z}^T \widehat{\boldsymbol{J}}_n^{\mathrm{BM}} \boldsymbol{z} = \sum_{e \in \widetilde{\mathcal{E}}} [\boldsymbol{W}_e \boldsymbol{D}_e \boldsymbol{z}]^T [\boldsymbol{W}_e \boldsymbol{z}].$$

These dot products involve different vectors and can be negative.

The symmetric part determines the sign of the BM quadratic form:

$$\boldsymbol{z}^T \widehat{\boldsymbol{J}}_n \boldsymbol{z} = \boldsymbol{z}^T \operatorname{sym}(\widehat{\boldsymbol{J}}_n) \boldsymbol{z}, \qquad \operatorname{sym}(\widehat{\boldsymbol{J}}_n) = \tfrac{1}{2}[\widehat{\boldsymbol{J}}_n + \widehat{\boldsymbol{J}}_n^T].$$

- **All eigenvalues nonnegative:** the quadratic form is nonnegative in every direction.
- **A negative eigenvalue:** the quadratic form is negative in at least one direction.

Classical ECSW preserves symmetry and real, nonnegative eigenvalues.

Nonnegative modal weights do not guarantee these properties for BM-ECSW.

These properties alone do not guarantee nonlinear convergence or trajectory accuracy.

Matrix calculations | Positive-eigenvalue example

With $N_c = 90$, BM fits to 1.28×10^{-15} and still fails online.

Same ROB ($n = 96$), $\tau = 10^{-5}$; N_c: training configurations; three tests of 500 steps.

Computational model	N_c	Cells n_e	$\lambda_{\min}/\lambda_{\max}$	$\mathbb{RE}^{\max}_{2,\boldsymbol{u}}$ (%)
HPROM (ECSW)	90	1 590	+0.678	1.096
HPROM (ECSW)	225	2 492	+0.678	1.096
HPROM (BM-ECSW)	90	145	−0.231	4 411.533
HPROM (BM-ECSW)	225	300	+0.046	1.102

Ratio: median $\lambda_{\min}/\lambda_{\max}$ of $\mathrm{sym}(\widehat{\boldsymbol{J}}_n)$ at each step's initial guess. Errors: against the HDM.

- **Slide 10:** both BM rules fail the common-objective test, including the accurate one.
- **Slide 16:** negative quadratic directions for BM with $N_c = 90$ at every step; none with 225.

Even a tightly solved step can be inaccurate: first-step BM errors are **30.626%** for $N_c = 90$ and **0.048%** for 225, against the PROM.

Same previous state; each model's projected equations solved below 10^{-11} relative residual.

A small training error does not ensure accurate online equations.

More training data helps here; the selected mesh and weights also change.

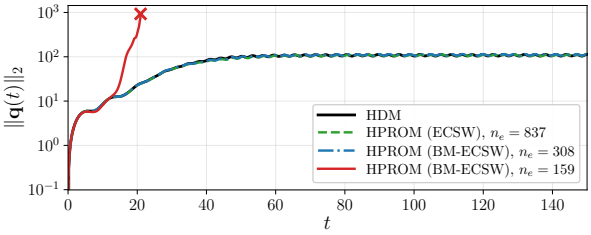
Diagnostics and trajectories Accurately solved steps

Cylinder flow: sensitivity to the training tolerance

AERO-F: unsteady 2D cylinder flow, Mach 0.2; Reynolds-parametric LSPG with $n = 96$.
Training: seven Reynolds numbers between 70 and 180, 385 snapshots. Test: $\mathrm{Re} = 100$.

Same training data; a tighter tolerance selects a larger sampled mesh.

Computed trajectories



HPROM (BM-ECSW)

	159 nodes	308 nodes
Training tolerance τ	10^{-2}	10^{-3}
$\lambda_{\min}/\lambda_{\max}$	-0.019	$+0.160$
Online result	Diverges at $t = 21$	Accurate to $t = 150$

Ratio: median $\lambda_{\min}/\lambda_{\max}$ of $\mathrm{sym}(\widehat{\boldsymbol{J}}_n)$ over assembled correction matrices (Slide 16).

With the same training data, the tighter fit produces an accurate trajectory.

The sampled mesh and weights also change; the comparison does not identify the cause of failure.

Full trajectories and qualifications

- **Substantial savings with comparable accuracy.** Successful Burgers case: $2\,492 \rightarrow 300$ **cells**, approximately $27.5\times$ **faster training** and $15.4\times$ **faster online computation**.
- **Additional flexibility changes the inherited guarantees.** Modal weighting does not automatically preserve one common LSPG objective, a symmetric positive semidefinite correction matrix, or the mechanical Lagrangian and energy conservation with the same reduced kinetic energy.
- **Training and online accuracy are separate checks.** The Burgers rule with $N_c = 90$ fits to 1.28×10^{-15} yet is inaccurate online; the $N_c = 225$ rule achieves comparable trajectory accuracy to classical ECSW.
- **Accurate online solutions remain possible.** The successful BM rule also lacks a common objective. The computations do not isolate which loss causes each failure.

BM offers substantial efficiency gains. Online validation and the structural requirements of the application remain essential when choosing a hyperreduction rule.

Competing proposals: $\boldsymbol{A}_e$ and grouped weights

New freedom: one matrix per cell, acting on its HDM residual components.

$$\boldsymbol{F}_A(\boldsymbol{q}) = \widetilde{\boldsymbol{r}}_n^A = \sum\nolimits_{e \in \widetilde{\mathcal{E}}} \boldsymbol{W}_e^T \boldsymbol{A}_e \boldsymbol{r}_e.$$

BM weights reduced equations after projection. Here, $\boldsymbol{A}_e$ weights the HDM residual before projection. $\boldsymbol{W}_e = \partial \boldsymbol{r}_e / \partial \boldsymbol{q}$; $\boldsymbol{A}_e = \boldsymbol{A}_e^T \succeq 0$ is trained offline and fixed online.

Burgers: two residual components per cell, with separate weights and one cross-weight.

$$\boldsymbol{A}_e \boldsymbol{r}_e = \begin{bmatrix} a_e & b_e \\ b_e & c_e \end{bmatrix} \begin{bmatrix} r_{u,e} \\ r_{v,e} \end{bmatrix} = \begin{bmatrix} a_e r_{u,e} + b_e r_{v,e} \\ b_e r_{u,e} + c_e r_{v,e} \end{bmatrix}.$$

- a_e, c_e: weights for the two HDM residual components.
- b_e: couples the u and v residuals **within the same cell**.

Three independent entries per cell, constrained by $a_e, c_e \geq 0$ and $b_e^2 \leq a_e c_e$.

The complete vector comes from one common objective:

$$\Phi_A(\boldsymbol{q}) = \tfrac{1}{2} \sum\nolimits_{e \in \widetilde{\mathcal{E}}} \boldsymbol{r}_e^T \boldsymbol{A}_e \boldsymbol{r}_e, \qquad \nabla_{\boldsymbol{q}} \Phi_A = \boldsymbol{F}_A.$$

Its Gauss–Newton correction matrix is symmetric positive semidefinite:

$$\widehat{\boldsymbol{J}}_n^A = \sum\nolimits_{e \in \widetilde{\mathcal{E}}} \boldsymbol{W}_e^T \boldsymbol{A}_e \boldsymbol{W}_e \succeq 0.$$

$\boldsymbol{A}_e = \xi_e \boldsymbol{I}$ recovers classical ECSW.

Grouped matrices · Back to conclusions

New freedom: cross-weights between residual components of different cells.

$$\boldsymbol{F}_{\mathcal{P}}(\boldsymbol{q}) = \widetilde{\boldsymbol{r}}_n^{\mathcal{P}} = \sum\nolimits_{P\in\mathcal{P}} \boldsymbol{W}_P^T \boldsymbol{A}_P \boldsymbol{R}_P.$$

$\boldsymbol{R}_P$ stacks the residuals in group P; $\boldsymbol{W}_P = \partial \boldsymbol{R}_P / \partial \boldsymbol{q}$.
Sampled groups are selected and matrices $\boldsymbol{A}_P = \boldsymbol{A}_P^T \succeq 0$ fitted offline; both are fixed online.

For a group of two cells, the matrix has the following blocks:

$$\boldsymbol{R}_P = \begin{bmatrix} \boldsymbol{r}_{e_1} \\ \boldsymbol{r}_{e_2} \end{bmatrix}, \qquad \boldsymbol{A}_P = \begin{bmatrix} \boldsymbol{A}_{P,11} & \boldsymbol{A}_{P,12} \\ \boldsymbol{A}_{P,12}^T & \boldsymbol{A}_{P,22} \end{bmatrix}.$$

- Diagonal blocks: weight and couple residual components **within each cell**.
- Off-diagonal blocks: couple residual components **between different cells**.

Setting $\boldsymbol{A}_{P,12} = \boldsymbol{0}$ recovers separate cell matrices.

These cross-weights also preserve one common objective:

$$\Phi_{\mathcal{P}}(\boldsymbol{q}) = \tfrac{1}{2} \sum\nolimits_{P\in\mathcal{P}} \boldsymbol{R}_P^T \boldsymbol{A}_P \boldsymbol{R}_P, \qquad \nabla_{\boldsymbol{q}} \Phi_{\mathcal{P}} = \boldsymbol{F}_{\mathcal{P}}.$$

$$\widetilde{\boldsymbol{J}}_n^{\mathcal{P}} = \sum\nolimits_{P\in\mathcal{P}} \boldsymbol{W}_P^T \boldsymbol{A}_P \boldsymbol{W}_P \succeq 0 \quad \text{(symmetric positive semidefinite)}.$$

Compare against ECSW and BM: sampled cells, training cost, online time and trajectory accuracy.
Experimental competitors: cost and mesh gains remain open. Physical energy conservation is not established.

◂ Individual matrices ◂ Back to conclusions

Affine trial space: fixed $\boldsymbol{u}_{\mathrm{ref}}$ and $\boldsymbol{V}$; unknown coordinates $\boldsymbol{q}$.

$$\widetilde{\boldsymbol{u}}(\boldsymbol{q}) = \boldsymbol{u}_{\mathrm{ref}} + \boldsymbol{V}\boldsymbol{q}, \qquad \boldsymbol{V} \in \mathbb{R}^{N\times n}, \qquad n \ll N.$$

With $\boldsymbol{r}(\boldsymbol{q}) := \boldsymbol{r}^{m+1}(\widetilde{\boldsymbol{u}}(\boldsymbol{q}))$, LSPG solves at each time step:

$$\boldsymbol{q}^\star \in \operatorname*{arg\,min}_{\boldsymbol{q}\in\mathbb{R}^n} \Phi(\boldsymbol{q}), \qquad \Phi(\boldsymbol{q}) = \frac{1}{2}\left\|\boldsymbol{r}(\boldsymbol{q})\right\|_2^2.$$

Common scalar objective: one residual-error measure for all coordinates.
To impose stationarity: differentiate the **same scalar objective** Φ with respect to every coordinate, and set all derivatives to zero.

$$\boldsymbol{r}_n(\boldsymbol{q}) = \nabla_{\boldsymbol{q}}\Phi(\boldsymbol{q}) = \boldsymbol{W}(\boldsymbol{q})^T\boldsymbol{r}(\boldsymbol{q}), \qquad \boldsymbol{r}_n(\boldsymbol{q}^\star) = \boldsymbol{0},$$

$$\boldsymbol{W}(\boldsymbol{q}) = \frac{\partial \boldsymbol{r}(\boldsymbol{q})}{\partial \boldsymbol{q}} = \boldsymbol{J}(\widetilde{\boldsymbol{u}}(\boldsymbol{q}))\boldsymbol{V}, \qquad \boldsymbol{J} = \frac{\partial \boldsymbol{r}^{m+1}}{\partial \boldsymbol{u}}.$$

Parameters and known history states are fixed during the solve.

From stationarity: seek $\nabla_{\boldsymbol{q}}\Phi(\boldsymbol{q}) = \boldsymbol{r}_n(\boldsymbol{q}) = \boldsymbol{0}$.
At iteration k, let $\boldsymbol{r}^k = \boldsymbol{r}(\boldsymbol{q}^k)$ and $\boldsymbol{W}^k = \boldsymbol{W}(\boldsymbol{q}^k)$.
Gauss–Newton: linearize the HDM residual around $\boldsymbol{q}^k$:

$$\boldsymbol{r}(\boldsymbol{q}^k + \Delta\boldsymbol{q}) \approx \boldsymbol{r}^k + \boldsymbol{W}^k\Delta\boldsymbol{q}.$$

Local quadratic model: minimize $\widehat{\Phi}_k(\Delta\boldsymbol{q}) = \frac{1}{2}\|\boldsymbol{r}^k + \boldsymbol{W}^k\Delta\boldsymbol{q}\|_2^2$ over $\Delta\boldsymbol{q}$.
Model stationarity: set $\nabla_{\Delta\boldsymbol{q}}\widehat{\Phi}_k = \boldsymbol{0}$ to obtain the coupled correction:

$$\underbrace{[\boldsymbol{W}^k]^T\boldsymbol{W}^k}_{\widehat{\boldsymbol{J}}_n^k}\Delta\boldsymbol{q}^k = -\underbrace{[\boldsymbol{W}^k]^T\boldsymbol{r}^k}_{\boldsymbol{r}_n(\boldsymbol{q}^k)},$$

Corrected coordinates: $\boldsymbol{q}^{k+1} = \boldsymbol{q}^k + \Delta\boldsymbol{q}^k$.

Note 1 — Correction matrix: symmetric positive semidefinite.

Note 2 — Gauss–Newton approximation: residual-weighted second derivatives are omitted.

Vector form: BM-ECSW weights each component of the projected contributions:

$$\widetilde{\boldsymbol{r}}_n^{\mathrm{BM}}(\boldsymbol{q}) = \sum_{e \in \widetilde{\mathcal{E}}} \boldsymbol{D}_e \boldsymbol{g}_e(\boldsymbol{q}), \qquad \boldsymbol{D}_e = \mathrm{diag}(\xi_{e,1}, \ldots, \xi_{e,n}).$$

Componentwise form ($i = 1, \ldots, n$): the scalar weight ξ_e becomes a separate weight $\xi_{e,i}$ for each mode.

Classical ECSW

One weight per entity ($\xi_e \geq 0$):

$$\widetilde{r}_{n,i}^{\mathrm{ECSW}}(\boldsymbol{q}) = \sum_{e \in \widetilde{\mathcal{E}}} \xi_e \, g_{e,i}(\boldsymbol{q}),$$

BM-ECSW

One weight per entity and mode ($\xi_{e,i} \geq 0$):

$$\widetilde{r}_{n,i}^{\mathrm{BM}}(\boldsymbol{q}) = \sum_{e \in \widetilde{\mathcal{E}}} \xi_{e,i} \, g_{e,i}(\boldsymbol{q}),$$

One shared mesh for all BM modes: keep an entity if at least one modal weight is positive.
Aim: fit the training contributions with fewer sampled entities.
Weights are fixed online. Setting $\xi_{e,i} = \xi_e$ for every mode recovers classical ECSW.

The local functions have continuous second derivatives. The sampled set and weights are fixed.

Classical ECSW

$$F_{\mathrm{ECSW},i} = \sum_{e\in\widetilde{\mathcal{E}}} \xi_e \frac{\partial \phi_e}{\partial q_i}.$$

Differentiating the two components gives

$$\frac{\partial F_{\mathrm{ECSW},i}}{\partial q_j} = \sum_{e\in\widetilde{\mathcal{E}}} \xi_e \frac{\partial^2 \phi_e}{\partial q_j \partial q_i},$$

$$\frac{\partial F_{\mathrm{ECSW},j}}{\partial q_i} = \sum_{e\in\widetilde{\mathcal{E}}} \xi_e \frac{\partial^2 \phi_e}{\partial q_i \partial q_j}.$$

Subtracting gives

$$\frac{\partial F_{\mathrm{ECSW},i}}{\partial q_j} - \frac{\partial F_{\mathrm{ECSW},j}}{\partial q_i} = \sum_{e\in\widetilde{\mathcal{E}}} \xi_e \underbrace{\left[\frac{\partial^2 \phi_e}{\partial q_j \partial q_i} - \frac{\partial^2 \phi_e}{\partial q_i \partial q_j}\right]}_{=0} = 0.$$

BM-ECSW

$$F_{\mathrm{BM},i} = \sum_{e\in\widetilde{\mathcal{E}}} \xi_{e,i} \frac{\partial \phi_e}{\partial q_i}.$$

Differentiating the two components gives

$$\frac{\partial F_{\mathrm{BM},i}}{\partial q_j} = \sum_{e\in\widetilde{\mathcal{E}}} \xi_{e,i} \frac{\partial^2 \phi_e}{\partial q_j \partial q_i},$$

$$\frac{\partial F_{\mathrm{BM},j}}{\partial q_i} = \sum_{e\in\widetilde{\mathcal{E}}} \xi_{e,j} \frac{\partial^2 \phi_e}{\partial q_i \partial q_j}.$$

The local derivatives agree; subtracting gives

$$\frac{\partial F_{\mathrm{BM},i}}{\partial q_j} - \frac{\partial F_{\mathrm{BM},j}}{\partial q_i} = \underbrace{\sum_{e\in\widetilde{\mathcal{E}}} [\xi_{e,i} - \xi_{e,j}] \frac{\partial^2 \phi_e}{\partial q_i \partial q_j}}_{\text{not necessarily zero}}.$$

◂ Back to weighting comparison ◂ Back to BM mechanics

With $\Phi_{\mathrm{ECSW}} = \sum_{e\in\widetilde{\mathcal{E}}} \xi_e \phi_e$ and constant symmetric positive definite $\boldsymbol{M}_r$,

$$\mathcal{T}_r = \tfrac{1}{2}\dot{\boldsymbol{q}}^T \boldsymbol{M}_r \dot{\boldsymbol{q}}, \qquad \widetilde{\mathcal{L}}_r^{\mathrm{ECSW}} = \mathcal{T}_r - \Phi_{\mathrm{ECSW}}.$$

Euler–Lagrange steps

1. **Differentiate in velocity:**

$$\frac{\partial \widetilde{\mathcal{L}}_r^{\mathrm{ECSW}}}{\partial \dot{\boldsymbol{q}}} = \boldsymbol{M}_r \dot{\boldsymbol{q}},$$

2. **Differentiate in position:**

$$\frac{\partial \widetilde{\mathcal{L}}_r^{\mathrm{ECSW}}}{\partial \boldsymbol{q}} = -\nabla_{\boldsymbol{q}} \Phi_{\mathrm{ECSW}} = -\boldsymbol{F}_{\mathrm{ECSW}}.$$

3. **Take the time derivative and substitute:**

$$\frac{d}{dt}\left[\frac{\partial \widetilde{\mathcal{L}}_r^{\mathrm{ECSW}}}{\partial \dot{\boldsymbol{q}}}\right] - \frac{\partial \widetilde{\mathcal{L}}_r^{\mathrm{ECSW}}}{\partial \boldsymbol{q}} = \boldsymbol{0}.$$

With constant mass, this becomes

$$\boldsymbol{M}_r \ddot{\boldsymbol{q}} + \boldsymbol{F}_{\mathrm{ECSW}} = \boldsymbol{0}.$$

Compute total energy from that same Lagrangian.

$$\begin{aligned}\widetilde{\mathcal{H}}_r &= \dot{\boldsymbol{q}}^T \frac{\partial \widetilde{\mathcal{L}}_r^{\mathrm{ECSW}}}{\partial \dot{\boldsymbol{q}}} - \widetilde{\mathcal{L}}_r^{\mathrm{ECSW}} \\ &= \dot{\boldsymbol{q}}^T \boldsymbol{M}_r \dot{\boldsymbol{q}} - [\tfrac{1}{2}\dot{\boldsymbol{q}}^T \boldsymbol{M}_r \dot{\boldsymbol{q}} - \Phi_{\mathrm{ECSW}}] \\ &= \mathcal{T}_r + \Phi_{\mathrm{ECSW}}.\end{aligned}$$

Differentiating along the motion gives

$$\begin{aligned}\frac{d\widetilde{\mathcal{H}}_r}{dt} &= \dot{\boldsymbol{q}}^T [\boldsymbol{M}_r \ddot{\boldsymbol{q}} + \nabla_{\boldsymbol{q}} \Phi_{\mathrm{ECSW}}] \\ &= \dot{\boldsymbol{q}}^T [-\boldsymbol{F}_{\mathrm{ECSW}} + \boldsymbol{F}_{\mathrm{ECSW}}] = 0.\end{aligned}$$

The potential generates the internal vector and makes the energy derivative zero.

No external loading; conservation in continuous time. Discrete conservation depends on the time integrator.

◂ Back to Lagrangian structure ◂ Back to BM mechanics

Keep both elements and their original potential energies:

$$\phi_1 = \mathcal{V}^{\text{int}}_{1,r} = \tfrac{1}{2}[q_1 + q_2]^2, \qquad \phi_2 = \mathcal{V}^{\text{int}}_{2,r} = \tfrac{1}{2}[q_1 + 2q_2]^2.$$

Differentiating gives the local internal vectors:

$$\nabla_{\boldsymbol{q}}\phi_1 = \begin{pmatrix} q_1 + q_2 \\ q_1 + q_2 \end{pmatrix}, \qquad \nabla_{\boldsymbol{q}}\phi_2 = \begin{pmatrix} q_1 + 2q_2 \\ 2[q_1 + 2q_2] \end{pmatrix}.$$

Classical ECSW: $\xi_1 = \xi_2 = 1$.

$$\begin{aligned} \boldsymbol{F}_{\text{ECSW}} &= \nabla_{\boldsymbol{q}}\phi_1 + \nabla_{\boldsymbol{q}}\phi_2 \\ &= \begin{pmatrix} 2q_1 + 3q_2 \\ 3q_1 + 5q_2 \end{pmatrix}. \end{aligned}$$

$$\frac{\partial F_{\text{ECSW},1}}{\partial q_2} = \frac{\partial [2q_1 + 3q_2]}{\partial q_2} = 3,$$

$$\frac{\partial F_{\text{ECSW},2}}{\partial q_1} = \frac{\partial [3q_1 + 5q_2]}{\partial q_1} = 3.$$

The common potential is $\Phi_{\text{ECSW}} = \phi_1 + \phi_2$.

BM-ECSW: $\boldsymbol{D}_1 = \text{diag}(1, 4)$, $\boldsymbol{D}_2 = \boldsymbol{I}_2$.

$$\begin{aligned} \boldsymbol{F}_{\text{BM}} &= \begin{pmatrix} q_1 + q_2 \\ 4[q_1 + q_2] \end{pmatrix} + \begin{pmatrix} q_1 + 2q_2 \\ 2[q_1 + 2q_2] \end{pmatrix} \\ &= \begin{pmatrix} 2q_1 + 3q_2 \\ 6q_1 + 8q_2 \end{pmatrix}. \end{aligned}$$

$$\frac{\partial F_{\text{BM},1}}{\partial q_2} = \frac{\partial [2q_1 + 3q_2]}{\partial q_2} = 3,$$

$$\frac{\partial F_{\text{BM},2}}{\partial q_1} = \frac{\partial [6q_1 + 8q_2]}{\partial q_1} = 6.$$

No common potential generates this vector.

The general difference gives $[1 - 4]\,1 + [1 - 1]\,2 = -3$, agreeing with $3 - 6$.

Effect on stability ◂ Back to mechanics

The same two-element example has local positive semidefinite stiffness matrices:

$$\boldsymbol{K}_{1,r} = \nabla^2_{\boldsymbol{q}}\phi_1 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \boldsymbol{K}_{2,r} = \nabla^2_{\boldsymbol{q}}\phi_2 = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}.$$

Classical ECSW: stable oscillatory modes.

$$\boldsymbol{K}_r^{\mathrm{ECSW}} = \boldsymbol{K}_{1,r} + \boldsymbol{K}_{2,r} = \begin{pmatrix} 2 & 3 \\ 3 & 5 \end{pmatrix}.$$

For every nonzero $\boldsymbol{z}$,

$$\boldsymbol{z}^T \boldsymbol{K}_r^{\mathrm{ECSW}} \boldsymbol{z} = [z_1 + z_2]^2 + [z_1 + 2z_2]^2 > 0.$$

Both squares vanish only for $\boldsymbol{z} = \boldsymbol{0}$, so the stiffness is positive definite.

BM-ECSW: a negative stiffness eigenvalue.

$$\boldsymbol{K}_r^{\mathrm{BM}} = \boldsymbol{D}_1 \boldsymbol{K}_{1,r} + \boldsymbol{K}_{2,r} = \begin{pmatrix} 2 & 3 \\ 6 & 8 \end{pmatrix}.$$

$$\det \boldsymbol{K}_r^{\mathrm{BM}} = 2 \cdot 8 - 3 \cdot 6 = -2,$$

$$\lambda_- = 5 - 3\sqrt{3} < 0.$$

For $\boldsymbol{M}_r = \boldsymbol{I}_2$, the motion $\boldsymbol{q} = a(t)\boldsymbol{v}_-$ along the negative-eigenvalue mode satisfies

$$\ddot{a} + \lambda_- a = 0, \qquad a(t) = C_+ e^{\sqrt{-\lambda_-}\,t} + C_- e^{-\sqrt{-\lambda_-}\,t}.$$

Positive modal weights introduce an exponentially growing mode into a stable classical model.

This is a structural counterexample; it does not establish the cause of the flow-computation failures.

◂ Back to potential test ◂ Back to mechanics

Differentiate the local squared residual norm.

$$\phi_e = \tfrac{1}{2} \boldsymbol{r}_e^T \boldsymbol{r}_e, \qquad \boldsymbol{W}_e = \frac{\partial \boldsymbol{r}_e}{\partial \boldsymbol{q}}.$$

For each coordinate q_i, the product rule gives

$$\frac{\partial \phi_e}{\partial q_i} = \tfrac{1}{2} \left[\left[\frac{\partial \boldsymbol{r}_e}{\partial q_i} \right]^T \boldsymbol{r}_e + \boldsymbol{r}_e^T \frac{\partial \boldsymbol{r}_e}{\partial q_i} \right] = \boldsymbol{W}_e(:, i)^T \boldsymbol{r}_e,$$

$$\nabla_{\boldsymbol{q}} \phi_e = \tfrac{1}{2} [\boldsymbol{W}_e^T \boldsymbol{r}_e + [\boldsymbol{r}_e^T \boldsymbol{W}_e]^T] = \boldsymbol{W}_e^T \boldsymbol{r}_e.$$

The two scalar terms in the component derivative are equal, canceling the factor $1/2$.

Full LSPG

$$\Phi = \sum_{e \in \mathcal{E}} \phi_e = \tfrac{1}{2} \|\boldsymbol{r}(\boldsymbol{q})\|_2^2.$$

$$\nabla_{\boldsymbol{q}} \Phi = \sum_{e \in \mathcal{E}} \nabla_{\boldsymbol{q}} \phi_e = \sum_{e \in \mathcal{E}} \boldsymbol{W}_e^T \boldsymbol{r}_e = \boldsymbol{r}_n = \boldsymbol{F}.$$

The residual blocks partition the assembled HDM residual rows.

Classical ECSW: fixed scalar weights

$$\Phi_{\mathrm{ECSW}} = \sum_{e \in \widetilde{\mathcal{E}}} \xi_e \phi_e = \tfrac{1}{2} \sum_{e \in \widetilde{\mathcal{E}}} \xi_e \|\boldsymbol{r}_e\|_2^2.$$

$$\nabla_{\boldsymbol{q}} \Phi_{\mathrm{ECSW}} = \sum_{e \in \widetilde{\mathcal{E}}} \xi_e \nabla_{\boldsymbol{q}} \phi_e = \sum_{e \in \widetilde{\mathcal{E}}} \xi_e \boldsymbol{W}_e^T \boldsymbol{r}_e = \widetilde{\boldsymbol{r}}_n^{\mathrm{ECSW}} = \boldsymbol{F}_{\mathrm{ECSW}}.$$

Its gradient generates every sampled projected equation.

◂ Back to LSPG objective

The local functions are $\phi_e = \frac{1}{2} r_e^2$, with $r_1 = q_1$, $r_2 = q_2$, and $r_3 = q_1 + q_2^2 - 3$. Their gradients supply the projected contributions:

$$\nabla_{\boldsymbol{q}} \phi_e = \boldsymbol{W}_e^T r_e, \qquad \boldsymbol{W}_1 = (1, 0), \quad \boldsymbol{W}_2 = (0, 1), \quad \boldsymbol{W}_3 = (1, 2q_2).$$

Classical ECSW

$$\widetilde{\boldsymbol{r}}_n^{\mathrm{ECSW}} = 2 \begin{pmatrix} r_1 \\ 0 \end{pmatrix} + 3 \begin{pmatrix} 0 \\ r_2 \end{pmatrix} + 2 \begin{pmatrix} r_3 \\ 2q_2 r_3 \end{pmatrix} = \begin{pmatrix} 2q_1 + 2r_3 \\ 3q_2 + 4q_2 r_3 \end{pmatrix}.$$

The scalar weights define one common objective:

$$\Phi_{\mathrm{ECSW}} = \tfrac{1}{2} \left[2q_1^2 + 3q_2^2 + 2r_3^2 \right].$$

Differentiating verifies both components:

$$\frac{\partial \Phi_{\mathrm{ECSW}}}{\partial q_1} = 2q_1 + 2r_3,$$

$$\frac{\partial \Phi_{\mathrm{ECSW}}}{\partial q_2} = 3q_2 + 4q_2 r_3.$$

BM-ECSW

$$\widetilde{\boldsymbol{r}}_n^{\mathrm{BM}} = 2 \begin{pmatrix} r_1 \\ 0 \end{pmatrix} + 3 \begin{pmatrix} 0 \\ r_2 \end{pmatrix} + \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} r_3 \\ 2q_2 r_3 \end{pmatrix} = \begin{pmatrix} 2q_1 + 2r_3 \\ 3q_2 + 2q_2 r_3 \end{pmatrix}.$$

The crossed derivatives give

$$\frac{\partial \widetilde{r}_{n,1}^{\mathrm{BM}}}{\partial q_2} - \frac{\partial \widetilde{r}_{n,2}^{\mathrm{BM}}}{\partial q_1} = 4q_2 - 2q_2 = 2q_2.$$

For $q_2 \neq 0$, this excludes every common smooth scalar objective locally.

Changing the weight pair of entity 3 from $(2, 1)$ to $(2, 2)$ recovers the classical vector and objective.

Back to LSPG example

The online target is the BM projected equation:

$$\widetilde{\boldsymbol{r}}_n^{\mathrm{BM}}(\boldsymbol{q}) = \sum\nolimits_{e\in\widetilde{\mathcal{E}}} \boldsymbol{D}_e \boldsymbol{W}_e(\boldsymbol{q})^T \boldsymbol{r}_e(\boldsymbol{q}) = \boldsymbol{0}.$$

With fixed weights, its exact Jacobian differentiates both $\boldsymbol{W}_e$ and $\boldsymbol{r}_e$:

$$\left[\boldsymbol{J}_n^{\mathrm{BM}}\right]_{ij} = \frac{\partial \widetilde{r}_{n,i}^{\mathrm{BM}}}{\partial q_j}$$

$$= \underbrace{\sum\nolimits_{e\in\widetilde{\mathcal{E}}} \xi_{e,i} \boldsymbol{W}_e(:,i)^T \boldsymbol{W}_e(:,j)}_{[\widehat{\boldsymbol{J}}_n^{\mathrm{BM}}]_{ij}} + \underbrace{\sum\nolimits_{e\in\widetilde{\mathcal{E}}} \xi_{e,i} \boldsymbol{r}_e^T \frac{\partial^2 \boldsymbol{r}_e}{\partial q_j \partial q_i}}_{\text{residual-weighted second derivatives}} .$$

Retaining only the first term gives the matrix used online:

$$\widehat{\boldsymbol{J}}_n^{\mathrm{BM}} = \sum\nolimits_{e\in\widetilde{\mathcal{E}}} \boldsymbol{D}_e \boldsymbol{W}_e^T \boldsymbol{W}_e, \qquad \widehat{\boldsymbol{J}}_n^{\mathrm{BM}} \Delta\boldsymbol{q}^{\mathrm{BM}} = -\widetilde{\boldsymbol{r}}_n^{\mathrm{BM}}.$$

This is an approximate Newton correction for the BM projected equations. The omission of residual-weighted second derivatives explains the Gauss–Newton-type designation; it does not establish a common least-squares objective.

Classical Gauss–Newton minimizes its quadratic model; a full step does not guarantee a decrease in the nonlinear objective. Gradient compatibility is tested with the exact Jacobian, not its first-derivative approximation.

◂ Back to online correction

With fixed diagonal D_e, the crossed BM entries are

$$\left[\widehat{J}_n^{\mathrm{BM}}\right]_{ij} = \sum\nolimits_{e\in\widetilde{\mathcal{E}}} \xi_{e,i}\, \boldsymbol{W}_e(:,i)^T \boldsymbol{W}_e(:,j),$$

$$\left[\widehat{J}_n^{\mathrm{BM}}\right]_{ji} = \sum\nolimits_{e\in\widetilde{\mathcal{E}}} \xi_{e,j}\, \boldsymbol{W}_e(:,j)^T \boldsymbol{W}_e(:,i).$$

The two local inner products agree, so subtraction gives

$$[\widehat{J}_n^{\mathrm{BM}}]_{ij} - [\widehat{J}_n^{\mathrm{BM}}]_{ji} = \sum\nolimits_{e\in\widetilde{\mathcal{E}}} [\xi_{e,i} - \xi_{e,j}]\, \boldsymbol{W}_e(:,i)^T \boldsymbol{W}_e(:,j).$$

The quadratic forms follow by grouping the products:

$$\boldsymbol{z}^T \widehat{J}_n^{\mathrm{ECSW}} \boldsymbol{z} = \sum\nolimits_{e\in\widetilde{\mathcal{E}}} \xi_e [\boldsymbol{W}_e \boldsymbol{z}]^T [\boldsymbol{W}_e \boldsymbol{z}] = \sum\nolimits_{e\in\widetilde{\mathcal{E}}} \xi_e \|\boldsymbol{W}_e \boldsymbol{z}\|_2^2 \geq 0,$$

$$\boldsymbol{z}^T \widehat{J}_n^{\mathrm{BM}} \boldsymbol{z} = \sum\nolimits_{e\in\widetilde{\mathcal{E}}} \boldsymbol{z}^T \boldsymbol{D}_e \boldsymbol{W}_e^T \boldsymbol{W}_e \boldsymbol{z} = \sum\nolimits_{e\in\widetilde{\mathcal{E}}} [\boldsymbol{W}_e \boldsymbol{D}_e \boldsymbol{z}]^T [\boldsymbol{W}_e \boldsymbol{z}].$$

A scalar equals its transpose. Averaging both forms gives

$$\begin{aligned} \boldsymbol{z}^T \widehat{J}_n \boldsymbol{z} &= \tfrac{1}{2}[\boldsymbol{z}^T \widehat{J}_n \boldsymbol{z} + \boldsymbol{z}^T \widehat{J}_n^T \boldsymbol{z}] \\ &= \boldsymbol{z}^T \operatorname{sym}(\widehat{J}_n) \boldsymbol{z}, \qquad \operatorname{sym}(\widehat{J}_n) = \tfrac{1}{2}[\widehat{J}_n + \widehat{J}_n^T]. \end{aligned}$$

For $\operatorname{sym}(\widehat{J}_n)\boldsymbol{v} = \lambda \boldsymbol{v}$, $\boldsymbol{v}^T \widehat{J}_n \boldsymbol{v} = \lambda \|\boldsymbol{v}\|_2^2$. Setting $\boldsymbol{D}_e = \xi_e \boldsymbol{I}_n$ recovers the classical matrix.

◂ Back to correction matrices

Hold the matrix fixed in the auxiliary model $\dot{\boldsymbol{z}} = -\widehat{\boldsymbol{J}}_n \boldsymbol{z}$.

$$\frac{d}{dt}\,\tfrac{1}{2}\|\boldsymbol{z}\|_2^2 = \boldsymbol{z}^T\dot{\boldsymbol{z}} = -\boldsymbol{z}^T\widehat{\boldsymbol{J}}_n\boldsymbol{z} = -\boldsymbol{z}^T\,\mathrm{sym}(\widehat{\boldsymbol{J}}_n)\boldsymbol{z}.$$

A positive quadratic form decreases the perturbation norm; a negative one initially increases it. Using $\boldsymbol{W}_1 = (1,0)$, $\boldsymbol{W}_2 = (0,1)$, and $\boldsymbol{W}_3 = (1,1)$ gives:

Classical ECSW

$$\boldsymbol{\xi} = \begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix}, \qquad \widehat{\boldsymbol{J}}_n^{\mathrm{ECSW}} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}.$$

The matrix is symmetric positive definite, with eigenvalues 1 and 9.
For $\boldsymbol{z} = (1,-2)^T$,

$$\boldsymbol{z}^T\widehat{\boldsymbol{J}}_n^{\mathrm{ECSW}}\boldsymbol{z} = 9, \qquad \frac{d}{dt}\,\tfrac{1}{2}\|\boldsymbol{z}\|_2^2\Big|_{t=0} = -9.$$

The perturbation norm decreases from the start.

BM-ECSW

$$\boldsymbol{\Xi} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 8 & 0 \end{pmatrix}, \qquad \widehat{\boldsymbol{J}}_n^{\mathrm{BM}} = \begin{pmatrix} 9 & 8 \\ 0 & 1 \end{pmatrix}.$$

The eigenvalues are again 1 and 9, but

$$\mathrm{sym}(\widehat{\boldsymbol{J}}_n^{\mathrm{BM}}) = \begin{pmatrix} 9 & 4 \\ 4 & 1 \end{pmatrix}, \quad \lambda = 5 \pm 4\sqrt{2}.$$

For the same $\boldsymbol{z}$, the quadratic form is -3, so $\frac{d}{dt}\,\frac{1}{2}\|\boldsymbol{z}\|_2^2\big|_{t=0} = +3$.

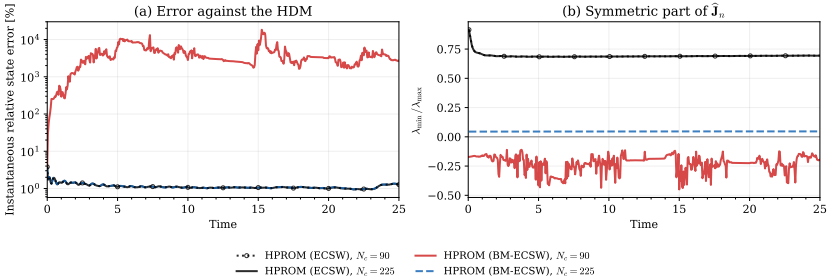
Both auxiliary models eventually decay; BM can amplify the perturbation temporarily.

The symmetric part has determinant -7 and one negative eigenvalue. The weights are nonnegative; entity 3 has zero weight in the second component. This auxiliary equation is not the HPROM physical evolution.

Back to correction matrices

The crossed derivatives are tested at every computed state: three parameters, 500 steps each.

Model	N_c	$\|\boldsymbol{J}_n - \boldsymbol{J}_n^T\|_F / \|\boldsymbol{J}_n\|_F$	Nonsymmetric
HPROM (ECSW)	90	8.95×10^{-17}	0/1 500
HPROM (ECSW)	225	8.74×10^{-17}	0/1 500
HPROM (BM-ECSW)	90	1.099	1 500/1 500
HPROM (BM-ECSW)	225	0.434	1 500/1 500



Trajectories at $\boldsymbol{\mu} = (4.56, 0.019)$; correction matrices at the first iterate of each step.

Back to Burgers

Test the equations separately from accumulated trajectory error.

Reference PROM: full-mesh residual, same 96-mode ROB; 20 steps at $\boldsymbol{\mu} = (4.56, 0.019)$.

At steps 1, 5, and 20, every model receives the same previous PROM state and solves its own projected equations to a relative residual below 10^{-11}.

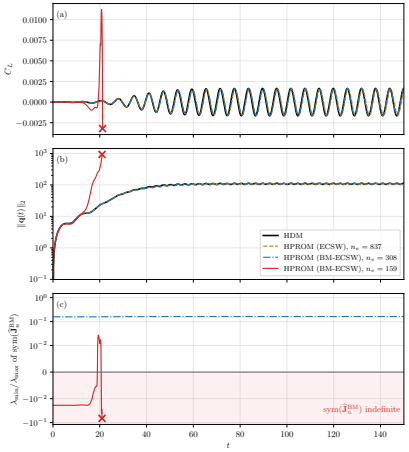
First-step results

Model	$\lambda_{\min}/\lambda_{\max}$	$\kappa_2(\widehat{\boldsymbol{J}}_n)$	State error [%]
PROM	$+0.915$	1.09	0
HPROM (ECSW), $N_c = 90$	$+0.915$	1.09	4.06×10^{-4}
HPROM (ECSW), $N_c = 225$	$+0.915$	1.09	9.17×10^{-6}
HPROM (BM-ECSW), $N_c = 90$	-0.172	3 233.39	30.626
HPROM (BM-ECSW), $N_c = 225$	$+0.044$	13.40	0.048

State errors are against the PROM; matrix ratios and condition numbers are at the common initial guess.

BM can satisfy its equations accurately and still produce an inaccurate state.
Failure of the nonlinear solver alone cannot explain the unsuccessful trajectory.

Back to Burgers



Same ROB ($n = 96$); test at $\mathrm{Re} = 100$.

159 nodes, $\tau = 10^{-2}$
Median symmetric-part ratio: -0.019; indefinite in 210 of 226 recorded matrices. The run stops at $t = 21$ with negative pressures.

308 nodes, $\tau = 10^{-3}$
Median ratio: $+0.160$; positive definite in all 1 500 recorded matrices. The trajectory remains close to the HDM and classical ECSW.

Additional check: eigenvalues of $\widehat{J}_n$.
Both rules have positive eigenvalue real parts at every recorded assembly, despite their different outcomes.

The ratio alone does not establish stability.
A 174-node BM rule diverges at $t = 41.5$, although this part is positive definite from $t = 25$ until failure.

Classical rules restricted to 159 and 174 nodes also diverge, at $t = 56$ and 80. A sufficiently small classical mesh can fail as well.

◂ Back to cylinder

Start with no sampled contributions:

$$k = 0, \qquad \mathcal{I}^0 = \varnothing, \qquad \mathcal{Z}^0 = \mathcal{E}, \qquad \boldsymbol{\xi}^0 = \mathbf{0}.$$

$\mathcal{I}^k$ contains the selected entities; $\mathcal{Z}^k = \mathcal{E} \setminus \mathcal{I}^k$ contains the remaining candidates.
Cubature fitting residual: measure the difference from the full target:

$$\boldsymbol{h}^k = \boldsymbol{b} - \boldsymbol{G}\boldsymbol{\xi}^k, \qquad \boldsymbol{h}^0 = \boldsymbol{b}.$$

Training accuracy: continue selection while

$$\eta^k = \frac{\|\boldsymbol{h}^k\|_2}{\|\boldsymbol{b}\|_2} > \tau.$$

After each refit, increase k and check the updated fitting error. The tolerance controls the accuracy of the cubature fit on the training data.

Back to algorithm

Training objective: $F(\boldsymbol{\xi}) = \frac{1}{2}\|\boldsymbol{G}\boldsymbol{\xi} - \boldsymbol{b}\|_2^2$.
Expand the quadratic:

$$\begin{aligned} 2F(\boldsymbol{\xi}) &= [\boldsymbol{G}\boldsymbol{\xi} - \boldsymbol{b}]^T[\boldsymbol{G}\boldsymbol{\xi} - \boldsymbol{b}] \\ &= \boldsymbol{\xi}^T\boldsymbol{G}^T\boldsymbol{G}\boldsymbol{\xi} - 2\boldsymbol{\xi}^T\boldsymbol{G}^T\boldsymbol{b} + \boldsymbol{b}^T\boldsymbol{b}. \end{aligned}$$

Differentiate with respect to the weights:

$$\nabla_{\boldsymbol{\xi}} F(\boldsymbol{\xi}^k) = \boldsymbol{G}^T\boldsymbol{G}\boldsymbol{\xi}^k - \boldsymbol{G}^T\boldsymbol{b} = -\boldsymbol{G}^T\boldsymbol{h}^k.$$

Hence the score is the negative fitting gradient:

$$\boldsymbol{\mu}^k = \boldsymbol{G}^T\boldsymbol{h}^k = -\nabla_{\boldsymbol{\xi}} F(\boldsymbol{\xi}^k), \qquad \boldsymbol{h}^k = \boldsymbol{b} - \boldsymbol{G}\boldsymbol{\xi}^k.$$

For column $\boldsymbol{g}_e = \boldsymbol{G}(:, e)$, $\mu_e^k = \boldsymbol{g}_e^T\boldsymbol{h}^k$. For an unselected entity, a positive score means that increasing its zero weight initially decreases F.

Back to algorithm

Feasible candidate: for $e \in \mathcal{Z}^k$, the current weight is zero. Increase it by $t \geq 0$, keeping the other weights fixed:

$$\boldsymbol{\xi}(t) = \boldsymbol{\xi}^k + t\boldsymbol{e}_e, \qquad \boldsymbol{G}\boldsymbol{e}_e = \boldsymbol{g}_e,$$

Here $\boldsymbol{e}_e$ is the unit vector for weight e, $\boldsymbol{h}^k = \boldsymbol{b} - \boldsymbol{G}\boldsymbol{\xi}^k$, and $\mu_e^k = \boldsymbol{g}_e^T \boldsymbol{h}^k$.
Effect on the fitting objective $F(\boldsymbol{\xi}) = \frac{1}{2}\|\boldsymbol{G}\boldsymbol{\xi} - \boldsymbol{b}\|_2^2$:

$$F(\boldsymbol{\xi}^k + t\boldsymbol{e}_e) = F(\boldsymbol{\xi}^k) - t\mu_e^k + \frac{t^2}{2}\|\boldsymbol{g}_e\|_2^2.$$

Compare initial slopes for the same weight increment:

$$\frac{d}{dt} F(\boldsymbol{\xi}^k + t\boldsymbol{e}_e)\Big|_{t=0} = -\mu_e^k, \qquad e^k \in \arg\max_{e \in \mathcal{Z}^k} \mu_e^k.$$

The largest positive score gives the greatest initial decrease. This initial-slope criterion does not guarantee the largest improvement after the NNLS refit.

Back to algorithm

Enlarge the working set and update the remaining candidates:

$$\mathcal{I}^{k+1} = \mathcal{I}^k \cup \{e^k\}, \qquad \mathcal{Z}^{k+1} = \mathcal{E} \setminus \mathcal{I}^{k+1}.$$

Refit all selected weights against the original target b:

$$\boldsymbol{\xi}^{k+1}_{\mathcal{I}^{k+1}} \in \arg\min_{\boldsymbol{z} \geq \mathbf{0}} \frac{1}{2} \| \boldsymbol{G}_{\mathcal{I}^{k+1}} \boldsymbol{z} - \boldsymbol{b} \|_2^2, \qquad \boldsymbol{\xi}^{k+1}_{\mathcal{Z}^{k+1}} = \mathbf{0}.$$

Existing weights may increase, decrease, or become zero. Every unselected entity retains zero weight.

Then measure the error of the new complete rule:

$$\boldsymbol{h}^{k+1} = \boldsymbol{b} - \boldsymbol{G}\boldsymbol{\xi}^{k+1}.$$

This updated fitting residual determines the next scores and the stopping test.

Back to algorithm

An NNLS refit can assign zero weight to a selected entity. The final sampled mesh therefore contains the positive-weight entities:

$$\widetilde{\mathcal{E}} = \{e \in \mathcal{E} : \xi_e^k > 0\} \subseteq \mathcal{I}^k.$$

Only these entities contribute to the sampled projected residual:

$$\widetilde{\boldsymbol{r}}_n^{\mathrm{ECSW}}(\boldsymbol{q}) = \sum_{e \in \widetilde{\mathcal{E}}} \xi_e^k \boldsymbol{g}_e(\boldsymbol{q}).$$

Return both parts of the cubature rule: the full weight vector $\boldsymbol{\xi}^k$ and its positive support $\widetilde{\mathcal{E}}$. The weights are fixed during the online simulation.
Mesh size is an output: $|\widetilde{\mathcal{E}}|$ follows from the trained rule.

◂ Back to algorithm

One shared working set, initially empty:

$$k = 0, \qquad \mathcal{I}^0 = \varnothing, \qquad \mathcal{Z}^0 = \mathcal{E}.$$

Initialize every mode with zero weights:

$$\boldsymbol{\xi}^{[i],0} = \mathbf{0}, \qquad \boldsymbol{h}^{[i],0} = \boldsymbol{b}^{[i]}, \qquad i = 1, \ldots, n.$$

At later iterations, $\boldsymbol{h}^{[i],k} = \boldsymbol{b}^{[i]} - \boldsymbol{G}^{[i]}\boldsymbol{\xi}^{[i],k}$.

Check one aggregate training error: continue selection while

$$[\eta^k]^2 = \frac{\sum_i \|\boldsymbol{h}^{[i],k}\|_2^2}{\sum_i \|\boldsymbol{b}^{[i]}\|_2^2} > \tau^2.$$

The denominator is positive by assumption. Individual modal targets may be zero; no separate relative tolerance is imposed on each mode.

◂ Back to BM algorithm

Expand the offline fitting objective: $\boldsymbol{\Xi} = [\boldsymbol{\xi}^{[1]} \ \cdots \ \boldsymbol{\xi}^{[n]}]$ collects the modal weights.

$$
\begin{aligned}
2F_{\mathrm{BM}}(\boldsymbol{\Xi}) &= \sum_i \|\boldsymbol{G}^{[i]}\boldsymbol{\xi}^{[i]} - \boldsymbol{b}^{[i]}\|_2^2 \\
&= \sum_i \Big[[\boldsymbol{\xi}^{[i]}]^T [\boldsymbol{G}^{[i]}]^T \boldsymbol{G}^{[i]}\boldsymbol{\xi}^{[i]} \\
&\qquad - 2[\boldsymbol{\xi}^{[i]}]^T [\boldsymbol{G}^{[i]}]^T \boldsymbol{b}^{[i]} + \|\boldsymbol{b}^{[i]}\|_2^2 \Big].
\end{aligned}
$$

Differentiate with respect to each mode's weights:

$$
\nabla_{\boldsymbol{\xi}^{[i]}} F_{\mathrm{BM}}\big|_{\boldsymbol{\Xi}^k} = [\boldsymbol{G}^{[i]}]^T [\boldsymbol{G}^{[i]}\boldsymbol{\xi}^{[i],k} - \boldsymbol{b}^{[i]}] = -[\boldsymbol{G}^{[i]}]^T \boldsymbol{h}^{[i],k}.
$$

The shared score sums these modal negative gradients:

$$
\boldsymbol{\mu}^k = \sum_i [\boldsymbol{G}^{[i]}]^T \boldsymbol{h}^{[i],k}.
$$

Here $\boldsymbol{h}^{[i],k} = \boldsymbol{b}^{[i]} - \boldsymbol{G}^{[i]}\boldsymbol{\xi}^{[i],k}$ is the fitting residual for mode i.

Why select the largest score?

◂ Back to BM algorithm

Compare a common increment in an unselected entity's modal weights:

$$\Xi(t) = \Xi^k + t\boldsymbol{e}_e\mathbf{1}_n^T, \qquad e \in \mathcal{Z}^k, \quad t \geq 0.$$

Here Ξ collects modal weights, $\boldsymbol{e}_e$ selects entity e, and $\mathbf{1}_n$ contains n ones.

Expand the offline fitting objective along this direction:

$$\begin{aligned} F_{\mathrm{BM}}(\Xi(t)) &= \frac{1}{2}\sum\nolimits_i \|\boldsymbol{h}^{[i],k} - t\boldsymbol{g}_e^{[i]}\|_2^2 \\ &= F_{\mathrm{BM}}(\Xi^k) - t\underbrace{\sum\nolimits_i [\boldsymbol{g}_e^{[i]}]^T\boldsymbol{h}^{[i],k}}_{\mu_e^k} + \frac{t^2}{2}\sum\nolimits_i \|\boldsymbol{g}_e^{[i]}\|_2^2. \end{aligned}$$

From the initial slope to the selection rule:

$$\left.\frac{d}{dt}F_{\mathrm{BM}}(\Xi(t))\right|_{t=0} = -\mu_e^k, \qquad e^k \in \arg\max_{e\in\mathcal{Z}^k} \mu_e^k.$$

Proposed heuristic: the algorithm is valid, but the score ranks equal modal increments. Signed correlations can cancel, so it may miss the largest gain after independent modal NNLS refits.

◂ Back to BM algorithm

Add one entity to the same working set for every mode:

$$\mathcal{I}^{k+1} = \mathcal{I}^k \cup \{e^k\}, \qquad \mathcal{Z}^{k+1} = \mathcal{E} \setminus \mathcal{I}^{k+1}.$$

For each mode, refit all selected weights against its original target:

$$\boldsymbol{\xi}^{[i],k+1}_{\mathcal{I}^{k+1}} \in \arg\min_{\boldsymbol{z}\geq \mathbf{0}} \frac{1}{2}\|\boldsymbol{G}^{[i]}_{\mathcal{I}^{k+1}}\boldsymbol{z} - \boldsymbol{b}^{[i]}\|_2^2.$$

These fits are independent once $\mathcal{I}^{k+1}$ is fixed. Previously selected weights may increase, decrease, or become zero.

Complete the weight vectors and update the modal fitting residuals:

$$\boldsymbol{\xi}^{[i],k+1}_{\mathcal{Z}^{k+1}} = \mathbf{0}, \qquad \boldsymbol{h}^{[i],k+1} = \boldsymbol{b}^{[i]} - \boldsymbol{G}^{[i]}\boldsymbol{\xi}^{[i],k+1}.$$

The updated residuals determine both the next shared selection and the aggregate stopping test. Zero-weight selected columns remain available for later refits.

◂ Back to BM algorithm

Take the union of the modal positive supports:

$$\widetilde{\mathcal{E}} = \bigcup_{i=1}^{n} \{e \in \mathcal{E} : \xi_{e,i}^{k} > 0\} \subseteq \mathcal{I}^{k}.$$

Keep an entity if at least one of its modal weights is positive. Omit it if all of its modal weights are zero.

Return one mesh and all modal weight vectors:

$$\{\boldsymbol{\xi}^{[i],k}\}_{i=1}^{n}, \qquad \widetilde{\mathcal{E}}.$$

They define the fixed online diagonal weights $\boldsymbol{D}_e = \mathrm{diag}(\xi_{e,1}^{k}, \ldots, \xi_{e,n}^{k})$.

Different weights, shared entities: the same sampled mesh supplies the contributions to every reduced equation.

◂ Back to BM algorithm

Appendix: online solution of HPROM (ECSW)

Input: Current time level t^{m+1}, prescribed parameters $\boldsymbol{\mu}$, known approximate history states on $\widetilde{\mathcal{E}}^+$, restricted basis $\boldsymbol{V}^+$, reference state $\boldsymbol{u}^+_{\text{ref}}$, selected mesh $\widetilde{\mathcal{E}}$, weights $\{\xi_e\}_{e\in\widetilde{\mathcal{E}}}$, initial coordinates $\boldsymbol{q}^{(0)}$, and tolerance $\varepsilon_{\text{NL}} \in (0,1)$

Output: Solution coordinates $\boldsymbol{q}^{m+1}$ at the current time step

$k \leftarrow 0$
loop
  $\widetilde{\boldsymbol{u}}^{+,(k)} \leftarrow \boldsymbol{u}^+_{\text{ref}} + \boldsymbol{V}^+\boldsymbol{q}^{(k)}$
  for all $e \in \widetilde{\mathcal{E}}$ **do**
    Evaluate $\boldsymbol{r}_e^{(k)}$ at the reconstructed state $\widetilde{\boldsymbol{u}}^{+,(k)}$
    $\boldsymbol{W}_e^{(k)} \leftarrow \frac{\partial \boldsymbol{r}_e}{\partial \boldsymbol{q}}(\boldsymbol{q}^{(k)})$
  end for
  $\widetilde{\boldsymbol{r}}_n^{\text{ECSW},(k)} \leftarrow \sum_{e\in\widetilde{\mathcal{E}}} \xi_e [\boldsymbol{W}_e^{(k)}]^T \boldsymbol{r}_e^{(k)}$
  if $k = 0$ **then**
    $\rho_0 \leftarrow \|\widetilde{\boldsymbol{r}}_n^{\text{ECSW},(0)}\|_2$
  end if
  if $\|\widetilde{\boldsymbol{r}}_n^{\text{ECSW},(k)}\|_2 \leq \varepsilon_{\text{NL}}\rho_0$ **then**
    return $\boldsymbol{q}^{m+1} \leftarrow \boldsymbol{q}^{(k)}$
  end if
  $\widehat{\boldsymbol{J}}_n^{\text{ECSW},(k)} \leftarrow \sum_{e\in\widetilde{\mathcal{E}}} \xi_e [\boldsymbol{W}_e^{(k)}]^T \boldsymbol{W}_e^{(k)}$
  Solve the reduced linear system for $\Delta\boldsymbol{q}^{(k)}$: $\widehat{\boldsymbol{J}}_n^{\text{ECSW},(k)} \Delta\boldsymbol{q}^{(k)} = -\widetilde{\boldsymbol{r}}_n^{\text{ECSW},(k)}$
  $\boldsymbol{q}^{(k+1)} \leftarrow \boldsymbol{q}^{(k)} + \Delta\boldsymbol{q}^{(k)}$
  $k \leftarrow k+1$
end loop

◂ Back to online comparison

Appendix: online solution of HPROM (BM-ECSW)

Input: Current time level t^{m+1}, prescribed parameters $\boldsymbol{\mu}$, known approximate history states on $\widetilde{\mathcal{E}}^{+}$, restricted basis $\boldsymbol{V}^{+}$, reference state $\boldsymbol{u}^{+}_{\mathrm{ref}}$, selected mesh $\widetilde{\mathcal{E}}$, modal weight matrices $\{\boldsymbol{D}_e\}_{e\in\widetilde{\mathcal{E}}}$, initial coordinates $\boldsymbol{q}^{(0)}$, and tolerance $\varepsilon_{\mathrm{NL}} \in (0,1)$

Output: Solution coordinates $\boldsymbol{q}^{m+1}$ at the current time step

$k \leftarrow 0$
loop
  $\widetilde{\boldsymbol{u}}^{+,(k)} \leftarrow \boldsymbol{u}^{+}_{\mathrm{ref}} + \boldsymbol{V}^{+}\boldsymbol{q}^{(k)}$
  for all $e \in \widetilde{\mathcal{E}}$ **do**
    Evaluate $\boldsymbol{r}_e^{(k)}$ at the reconstructed state $\widetilde{\boldsymbol{u}}^{+,(k)}$
    $\boldsymbol{W}_e^{(k)} \leftarrow \frac{\partial \boldsymbol{r}_e}{\partial \boldsymbol{q}}(\boldsymbol{q}^{(k)})$
  end for
  $\widetilde{\boldsymbol{r}}_n^{\mathrm{BM},(k)} \leftarrow \sum_{e\in\widetilde{\mathcal{E}}} \boldsymbol{D}_e[\boldsymbol{W}_e^{(k)}]^T \boldsymbol{r}_e^{(k)}$
  if $k = 0$ **then**
    $\rho_0 \leftarrow \|\widetilde{\boldsymbol{r}}_n^{\mathrm{BM},(0)}\|_2$
  end if
  if $\|\widetilde{\boldsymbol{r}}_n^{\mathrm{BM},(k)}\|_2 \le \varepsilon_{\mathrm{NL}}\rho_0$ **then**
    return $\boldsymbol{q}^{m+1} \leftarrow \boldsymbol{q}^{(k)}$
  end if
  $\widehat{\boldsymbol{J}}_n^{\mathrm{BM},(k)} \leftarrow \sum_{e\in\widetilde{\mathcal{E}}} \boldsymbol{D}_e[\boldsymbol{W}_e^{(k)}]^T \boldsymbol{W}_e^{(k)}$
  Solve the reduced linear system for $\Delta\boldsymbol{q}^{(k)}$: $\widehat{\boldsymbol{J}}_n^{\mathrm{BM},(k)}\Delta\boldsymbol{q}^{(k)} = -\widetilde{\boldsymbol{r}}_n^{\mathrm{BM},(k)}$
  $\boldsymbol{q}^{(k+1)} \leftarrow \boldsymbol{q}^{(k)} + \Delta\boldsymbol{q}^{(k)}$
  $k \leftarrow k+1$
end loop

◂ Back to online comparison